Runcheng Zhang

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EDUCATION

Nanjing University Sep 2025 – Present
B.S. in Intelligent Science and Technology Nanjing, China

RESEARCH EXPERIENCE

Tsinghua University – Natural Language Processing Lab (THUNLP)

Mar 2026 – Present

Research Intern, AI for Science | *Advisor: Zheni Zeng*

- Investigating language-model inference and sampling strategies for multi-objective molecular optimization, linking candidate generation to molecular property evaluation and scientific constraints.
- Prototyping scientific agents that combine inexpensive surrogate predictions, selective high-cost verification, executable analysis tools, and evidence-traceable hypotheses.

Microsoft Research Asia (Beijing)

Jan – Aug 2026

Research Intern, AI for Mathematics | *Advisor: Xian Zhang; Collaborator: Ziyu Zhou*

- Explored formal reasoning and autoformalization with Lean and Rzk, emphasizing native verifier feedback, typed proof obligations, error localization, and structured repair.
- Developed reproducible evaluation and proof-search utilities to separate verified proof progress from incomplete or invalid model outputs.

NJU Institute of Intelligent Networks and Communications (NINE)

Aug 2026 – Present

Research Intern | *Supervisor: Huihong Liang; Institute Director: Kun Yang*

- Studied efficient semantic video processing on constrained devices, including selective vision-token updates, temporal feature reuse, and reconstruction fidelity at narrow representation bottlenecks.
- Investigated CPU-native low-bit inference on ARM platforms and checked computational equivalence against reference implementations.

Wuxi Xishan NJU Institute of Applied Biotechnology

Sep 2026 – Present

Research Intern, Computational Enzyme Discovery

- Contributed to an enzyme–reaction discovery workflow connecting biochemical reaction queries, domain-specific evidence, and interactive computational tools for experimental research.

MANUSCRIPT

Input-Output Sampling for Efficient Multi-Objective Molecular Optimization with LLMs

Haoran Liu*, **Runcheng Zhang***, Zheni Zeng, Yukun Yan, Zhenghao Liu.

Submitted to ICLR 2027. *Equal contribution.

- Proposed MolDAS for multi-objective molecular RL: capability-aware sampling of diverse, learnable inputs; selective oracle feedback on low-probability output segments; and reward-dispersion-guided imitation with adaptive objective weights.
- In simulator-to-oracle training, improved optimization scores over vanilla GRPO by 3.7% in the low-fidelity stage and over 16.1% under capped high-fidelity calls, while reducing structural redundancy and training fluctuations.

PROFESSIONAL EXPERIENCE

Beijing Volcano Engine Technology Co., Ltd.

Aug 2026 – Present

Agent Development Engineer | *Client: Guizhou Xijiu Group*

- Developed components of a document-centric enterprise agent that structures multi-step analysis, draft generation, semantic review, and revision as tool-augmented workflows with explicit intermediate state.
- Integrated enterprise knowledge retrieval, model orchestration, and office-document tools through modular adapters; incorporated source-grounded checks to preserve verifiable facts across generated documents.

Nanjing Flux Technology Co., Ltd.

Jan 2026 – Present

Project Team Member | *Client-facing software development*

- Participate in backend and full-stack development, integrating reusable services, APIs, and model-assisted workflows within team-delivered projects.
- **Bank of Nanjing (outsourced proof of concept, Jan–Mar 2026):** implemented parts of a multimodal customer-emotion recognition pipeline combining speech preprocessing, speaker-aware transcription, and audio-language inference. Work remained at the PoC stage.

SELECTED RESEARCH PROJECTS

Evidence-Grounded Enzyme–Reaction Discovery (iGEM 2026)

2026

NJU-China Dry Lab | *Dry Lab Student Lead* | *Poster*

- Built a two-direction protein–reaction ranking framework that combines sequence and reaction representations with evidence-dependent expert corrections, without restricting predictions to expert-retrieved candidates.
- Connected ranking outputs to a scientific-agent workflow spanning reaction/sequence queries, graph-based evidence exploration, and explicit separation of verified knowledge from inferred candidates.
- Coordinated computational modeling and evaluation with wet-lab objectives, focusing on how researchers can inspect candidate provenance before experimental follow-up.

TRACE: Typed Reasoning with Attribution, Certificates, and Evidence

2026

Formal reasoning research linked to MSRA work | *Poster*

- Explored typed obligation graphs as a common interface between model-generated reasoning steps and independent native checkers for formal proof tasks.
- Organized certificate-backed conclusions, missing-support diagnoses, and replayable tool traces so unresolved search is not confused with a logically established answer.

Tangut-SPAR: Structured Proxy Augmentation and Reliable Preference Tuning

2026

Low-resource language modeling | *Poster*

- Studied data scarcity in Tangut translation using structured proxy augmentation to supply additional training signals while retaining linguistic constraints.
- Explored preference-based refinement of translation candidates, emphasizing faithful generation and the reliability of weak or proxy-derived supervision.

BiSem-Edge: Compute-Efficient Semantic Video

2026

NINE Lab | *Poster*

- Investigated selective visual-token refresh and cached temporal representations to avoid redundant encoder computation on largely unchanged frames.
- Analyzed information lost in low-dimensional semantic features and explored complementary reconstruction and CPU-native bitwise inference without conflating encoder speed with restored video quality.

SELECTED RESEARCH PROJECTS (CONTINUED)

Doge-SAGE: Stateful Agent Systems for Scientific Communities

2026

Agent infrastructure and scientific computing | *Poster*

- Designed a persistent, multi-user scientific assistant with shared capability discovery, turn-scoped state assembly, and event-local execution paths.
- Isolated concurrent task execution and merged committed history into durable group state, while applying admission checks before model or tool calls.

ADDITIONAL RESEARCH POSTERS

Exchangeable Hard Negatives for Vision–Language Evaluation 2026
Removed candidate-set structural shortcuts to test whether vision–language models use image evidence.

ProCo: Verifiable Abstention through Missing-Support Supervision 2026
Trained abstention with verifier-checkable witnesses of missing evidence in open-world reasoning.

Attributing Inference-Time Gains in Lean Theorem Proving 2026
Separated search-policy gains from retrieval, compute-budget, and prover-specific effects.

Structural Transfer in Low-Resource Autoformalization 2026
Diagnosed Rzk proof construction across boundaries, dependent lifts, and nested witnesses.

MOSAIC: Reference-Grounded Compositional Emoji Fusion 2025
Proposed source-aware visual composition with explicit roles, attribute transfer, and region planning.

INDEPENDENT MATHEMATICAL WRITING

Informal Notes on Mathematics

Ongoing

Author and editor of a multi-volume, English-language LaTeX collection of informal mathematical notes and essays. Rebuilt earlier multilingual material into a coherent book with thematic organization, corrected arguments, diagrams, and expanded explanations spanning algebra, geometry, topology, analysis, and logic. Project Notes library.

TECHNICAL PRESENTATIONS

Navigating the Full-Stack Ecosystem: Implementation Strategies and Knowledge Graphs for Information Aggregation Dec 2025

Student presenter, Intelligent Data and Decision Studio, Nanjing University. Presentation record.

Architectural Paradigms in Collaborative Software Engineering: A Comprehensive Analysis of the Nova Project Nov 2025

Student presenter, Intelligent Data and Decision Studio, Nanjing University. Presentation record.

TECHNICAL BACKGROUND

Research: LLM reasoning and scientific agents; multimodal representation; molecular optimization; formal verification; evaluation design.

Engineering: Python, TypeScript, PyTorch, Linux, Git; model/tool orchestration, knowledge retrieval, asynchronous services, resource-constrained inference.